

STA347 — Probability

Course Notes

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Lecture 1: Set Theory Fundamentals

Sep 8, 2026

“Probability = Set Theory + Measure”

1. Basic Set Operations

Union, Intersect, Complement, . . .

$A \cup B$, $A \cap B$, A^C , . . .

2. Def 1.1.1 Convergence of Real-Valued Sequences

A sequence $(x_n)_{n=1}^{\infty} \subseteq \mathbb{R}$ converges to x if

$$\forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ s.t. } n > N \Rightarrow |x_n - x| < \varepsilon.$$

$x_n \rightarrow x$ *converge to x?*

Pick ε (how close x_n is to the target).

For any ε picked, can find N , such that $n > N$, $|x_n - x| < \varepsilon$.

e.g. $x_n = \frac{1}{n} \rightarrow 0$

3. Def 1.1.2 Functional Convergence Pointwisely

A sequence $(f_n)_{n=1}^{\infty} \subseteq \mathbb{R}^X$ converges pointwise to f (*target f*) if

$$\forall \varepsilon > 0, \forall x \in X, \exists N_x \in \mathbb{N} \text{ s.t. } n > N_x \Rightarrow |f_n(x) - f(x)| < \varepsilon.$$

$f_1, \dots, f_n, \dots$ — *sequence of functions*

f — *target function*

For any ε and x , can find N_x s.t. $n > N_x, |f_n(x) - f(x)| < \varepsilon$.

$f_1(2), f_2(2), \dots, f_n(2), \dots \rightarrow f(2)$ — *for every x , there is a corresponding N_x .*

4. **Def 1.1.3 Functional Convergence Uniformly**

A sequence $(f_n)_{n=1}^\infty \subseteq \mathbb{R}^X$ converges uniformly to f if

$$\forall \varepsilon > 0, \exists N \in \mathbb{N} \quad \forall x \in X, n > N \Rightarrow |f_n(x) - f(x)| < \varepsilon.$$

That works for all $x \in X$ — need a common threshold N for uniform convergence.

Key difference:

For pointwise convergence, you need to have x specified first, then there is a threshold N_x corresponding to each x .

For uniform convergence, you need a common threshold N that works for all $x \in X$.

Example 1

$$f_n(x) = \frac{x}{n+x}, \quad x \geq 0$$

Then for any fixed value x , as $n \rightarrow \infty$, $f_n(x) \rightarrow 0$:

$$x = 1: f_n(1) = \frac{1}{n+1} \rightarrow 0 \text{ as } n \rightarrow \infty.$$

$$x = 2: f_n(2) = \frac{2}{n+2} \rightarrow 0 \text{ as } n \rightarrow \infty.$$

$$\therefore f_n(x) \xrightarrow{\text{pointwise}} 0$$

However, f_n does not converge uniformly to 0: no N works for all x .

No matter how you specify N , there is always a larger value of x for which the bound fails.

$$\Rightarrow f_n(x) \not\xrightarrow{\text{uniformly}} 0$$

Limiting Processes

Def 1.2.1 The Supremum of a Set

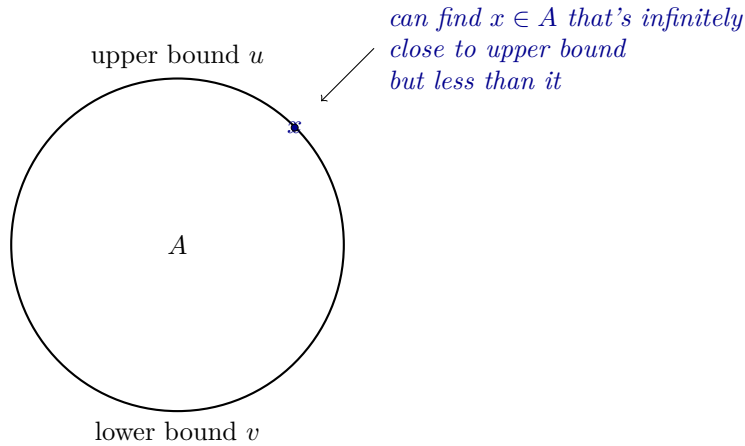
Let $A \subseteq \mathbb{R}$ be a subset. $u = \sup A$ iff

$$\forall \varepsilon > 0, \exists x \in A \quad u - x < \varepsilon \quad \text{and} \quad \forall x \in A, x \leq u.$$

Def 1.2.2 The Infimum of a Set

Let $A \subseteq \mathbb{R}$ be a subset. $v = \inf A$ iff

$$\forall \varepsilon > 0, \exists x \in A \quad x - v < \varepsilon \quad \text{and} \quad \forall x \in A, x \geq v.$$



Example 2

$$A = (0, 1)$$

$$\sup A = 1 \quad \inf A = 0$$

Distinction of sup/inf vs. max/min:

$\max(A)$ and $\min(A)$ must be elements of the set A .

$\sup(A)$ and $\inf(A)$ may not be elements of A .

Example 3

$$A = (0, 1)$$

$\max(A)$ and $\min(A)$ do not exist. $\sup(A)$ and $\inf(A)$ do exist.

$$B = (0, 1]$$

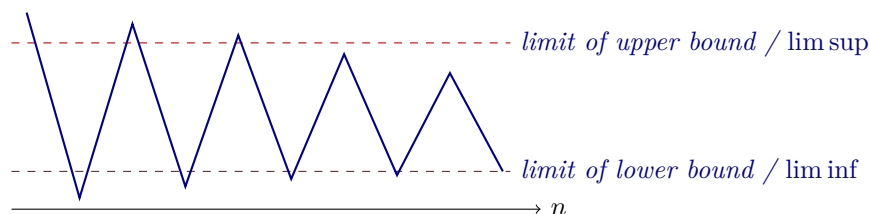
$\min(B)$ does not exist, $\max(B) = 1$. $\sup(B) = 1$, $\inf(B) = 0$.

Def 1.2.3 Limsup and Liminf

Let $(x_n) \subseteq \mathbb{R}$ be a sequence.

$$\limsup_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} \sup_{m \geq n} x_m = \inf_{n \geq 0} \sup_{m \geq n} x_m$$

$$\liminf_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} \inf_{m \geq n} x_m = \sup_{n \geq 0} \inf_{m \geq n} x_m$$



Example 4

$$x_n = (-1)^n = -1, 1, -1, 1, \dots$$

No limit, diverges: $\lim_{n \rightarrow \infty} x_n$ DNE.

But $\sup_{m \geq n} x_m = 1$ and $\inf_{m \geq n} x_m = -1$

$$\Rightarrow \limsup_{n \rightarrow \infty} x_n = 1, \quad \liminf_{n \rightarrow \infty} x_n = -1$$

The limsup and liminf are used to describe the long-run behaviour of a sequence, even if the sequence does not converge.

For a sequence x_n , if $\limsup_{n \rightarrow \infty} x_n = \liminf_{n \rightarrow \infty} x_n = x$, then $\lim_{n \rightarrow \infty} x_n = x$.

Function Limits**Def 1.3 Left and Right Limit of a Function**

For any function $f : \mathbb{R} \rightarrow \mathbb{R}$,

$$f(x-) = \lim_{y \uparrow x} f(y) = L \quad (\text{to the left of } x, \text{ a small neighbourhood})$$

$$\Leftrightarrow \forall \varepsilon > 0, \exists \delta > 0 \text{ s.t. } \forall y \in (x - \delta, x), |f(y) - L| < \varepsilon.$$

$$f(x+) = \lim_{y \downarrow x} f(y) = U$$

$$\Leftrightarrow \forall \varepsilon > 0, \exists \delta > 0 \text{ s.t. } \forall y \in (x, x + \delta), |f(y) - U| < \varepsilon.$$

If $f(x-) = f(x+) = C$, then $\lim_{y \rightarrow x} f(y) = C$.

Example 5

$$f(x) = \begin{cases} 0 & x < 0 \\ 1 & x \geq 0 \end{cases}$$

$$f(0-) = 0, f(0+) = 1$$

$$\Rightarrow \lim_{x \rightarrow 0} f(x) \text{ DNE}$$

Lecture 2: Probability Measure

Sep 10, 2026

Def 2.1 Sample Space

A sample space Ω is any non-empty set.

Example 6 The sample space contains "all the states of the following"

Coin Flipping: $\Omega : \{H, T\}$

Weather: $\{\text{hot, cold, mild}\} \times \{\text{wet, dry}\}$

Def 2.2 Probability Measure (STA347 ver.)

For a sample space Ω , we say P is a probability measure if the following holds:

- i) $\forall E \subseteq \Omega, 0 \leq P(E) \leq 1$, that is $P : \mathcal{P}(\Omega) \rightarrow [0, 1]$
- ii) $P(\Omega) = 1$
- iii) $\forall E_1, E_2, \dots \subseteq \Omega$ such that $E_i \cap E_j = \emptyset \forall i \neq j$

$$P\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} P(E_i)$$

$\mathcal{P}(\Omega)$ is the power set of Ω .

Remark: It may be impossible for such a P to exist. To define exactly which subsets we want P to satisfy these axioms, that would require more advanced mathematical theories. For this class, we just assume such a P is defined for all subsets of Ω .

Example 7 Probabilities of coin flipping

$$\Omega = \{H, T\}$$

$$\mathcal{P}(\Omega) = \{\emptyset, \{H\}, \{T\}, \{H, T\}\}$$

$$P : \mathcal{P}(\Omega) \mapsto [0, 1]$$

$$p : \emptyset \mapsto 0$$

$$\{H\} \mapsto \frac{1}{2}$$

$$\{T\} \mapsto \frac{1}{2}$$

$$\{H, T\} \mapsto 1$$

Example 8 The uniform/Lebesgue measure. Let $\Omega = [L, U]$ for some $L < U \in \mathbb{R}$. Define

$$P([a, b]) = \frac{b - a}{U - L} \quad \text{when } L \leq a \leq b \leq U$$

We assume such a measure exists (and is well defined) for this class.

Proposition 2.3: The following properties are satisfied by any probability measure P

1. $\forall E \subseteq \Omega, P(E^C) = 1 - P(E)$
2. $P(\phi) = 0$
3. $\forall E, F \subseteq \Omega$, such that $E \subseteq F, P(E) \leq P(F)$ (*“big event, big probability”*)
4. $\forall E, F \subseteq \Omega, P(E \cup F) = P(E) + P(F) - P(E \cap F)$
5. $\forall E, F \subseteq \Omega, P(E \cap F) = P(E) + P(F) - P(E \cup F)$
6. $\forall E, F \subseteq \Omega, P(F \setminus E) = P(F) - P(E \cap F)$

Proof via Def 2.2 (probability axioms).

Lemma 2.4

For a sequence $E_1, \dots, E_n \subseteq \Omega$

$$P\left(\bigcup_{i=1}^n E_i\right) \leq \sum_{i=1}^n P(E_i)$$

Proof: We define $F_i = E_i \setminus \bigcup_{j=1}^{i-1} E_j, F_1 = E_1$

$$\bigcup_{i=1}^n E_i = \bigcup_{i=1}^n F_i \text{ and } F_i \cap F_j = \phi \text{ if } i \neq j$$

from the probability axiom.

$$\begin{aligned} \text{Since } \bigcup_{i=1}^n E_i &= \bigcup_{i=1}^n F_i \\ P\left(\bigcup_{i=1}^n E_i\right) &= P\left(\bigcup_{i=1}^n F_i\right) \\ &= \sum_{i=1}^n P(F_i) \end{aligned}$$

We also know $F_i \subseteq E_i$ as $F_i = E_i \setminus \bigcup_{j=1}^{i-1} E_j$

$$\begin{aligned} P(F_i) &\leq P(E_i) \\ P\left(\bigcup_{i=1}^n E_i\right) &= \sum_{i=1}^n P(F_i) \leq \sum_{i=1}^n P(E_i) \end{aligned}$$

Lemma 2.5

Let P be the Lebesgue measure on $[0, 1]$. Then $P(E) = 0$ for any countable set E .

Lecture 3: Independence, Random Variables and Distributions

Sep 15, 2026

Def 2.7 Independence

Events $E_1, \dots, E_n \subseteq \Omega$ are independent with respect to a probability measure P , if for each $\mathcal{I} \subseteq \begin{matrix} [n] \\ \text{any subset of } \{1, \dots, n\} \end{matrix}$

$$P\left(\bigcap_{i \in \mathcal{I}} E_i\right) = \prod_{i \in \mathcal{I}} P(E_i)$$

e.g. if $\mathcal{I} = \{1, 2, 3\}$ a subset of $\{1, \dots, n\}$

$$P(E_1 \cap E_2 \cap E_3) = P(E_1)P(E_2)P(E_3)$$

Proposition 2.7.1 If $E_1, \dots, E_n$ are independent, then $E_1^c, \dots, E_n^c$ are also independent. Proof as an exercise.

Proposition 2.7.2 An infinite collection of events $\{E_\alpha : \alpha \in \mathcal{I}\}$ are independent. Then for a finite subset $\mathcal{J} \subseteq \mathcal{I}$, the events $\{E_\beta : \beta \in \mathcal{J}\}$ are independent.

Random Variables

Def. 3.1 Random Variables

Given a sample space Ω , a function $X : \Omega \rightarrow \mathbb{R}$ is a random variable.

Example 3.1.1 Coin Flipping

$$\Omega = \{H, T\}$$

$$\downarrow \text{R.V. } X : \quad X(H) \rightarrow 1, \quad X(T) \rightarrow 0$$

$$\mathbb{R}$$

Theorem 3.2 Distribution

A random variable X and probability measure P induce a probability measure μ on R defined by

$$\mu(A) = p(X \in A) = p(\{\omega \in \Omega : X(\omega) \in A\})$$

for any $A \subseteq R$ *μ is a measure on the real number set: $\mu(\{0, 1\})$ and $\mu([0, 1])$ each return a certain value.*

$\mu(A)$ takes a real number set A as input, and equals P of the *pre-image of A w.r.t. r.v. X* , i.e. $P(\Omega_A) = \mu(A)$ where $\Omega_A = \{\omega \in \Omega : X(\omega) \in A\}$.

Example 3.2.1 Define the following

$$\Omega = \{\text{blue}, \text{red}, \text{green}, \text{yellow}\}$$

$$X : \begin{cases} \text{blue} \rightarrow 1 \\ \text{red} \rightarrow 3 \\ \text{green} \rightarrow 4 \\ \text{yellow} \rightarrow 6 \end{cases}$$

$$P : \{\text{blue}\} \mapsto \frac{1}{5}, \quad \{\text{red}\} \mapsto \frac{1}{5}, \quad \{\text{green}\} \mapsto \frac{1}{4}, \quad \{\text{yellow}\} \mapsto 1 - \frac{1}{5} - \frac{1}{5} - \frac{1}{4} = \frac{7}{20}$$

$$P(\{\text{blue}, \text{red}\}) = \frac{1}{5} + \frac{1}{5} = \frac{2}{5}$$

$$A = [1, 3]$$

1. $\{w : w \in \Omega, X(w) \in A\} = \{\text{blue}, \text{red}\}$
2. $P(\{\text{blue}, \text{red}\}) = \frac{2}{5}$, so $\mu(A) = \frac{2}{5}$

in summary

$$\mu([1, 3]) = \frac{2}{5}$$

Claim: μ is a probability measure on $\mathbb{R}$ (for any real number set B).

proof:

1. $\mu(B) \in [0, 1]$ is trivial, since P is a probability measure.
2. Since $X(\omega) \in R$ for all $\omega \in \Omega$.

$$\mu(R) = p(\{\omega : \omega \in \Omega, X(\omega) \in R\}) = p(\Omega) = 1$$

3. Consider disjoint sets $A_1, A_2, \dots \subseteq R$ and define $E_i = \{\omega : X(\omega) \in A_i\}$
(e.g. $E_1 = \{\omega : X(\omega) \in A_1\}$, $E_2 = \{\omega : X(\omega) \in A_2\}$)

Since X is a function, it is impossible for $X(\omega) = a$ and $X(\omega) = b$ when $a \neq b$, so the E_i 's are also disjoint. Then

$$\begin{aligned} \mu\left(\bigcup_{i=1}^{\infty} A_i\right) &= p\left(X(\omega) \in \bigcup_{i=1}^{\infty} A_i\right) \\ &= p\left(\bigcup_{i=1}^{\infty} E_i\right) \quad \text{we know } E_i\text{'s are disjoint} \\ &= \sum_{i=1}^{\infty} p(E_i) \\ &= \sum_{i=1}^{\infty} \mu(A_i) \end{aligned}$$

□

Def. 3.3 The Cumulative Distribution Function

A random variable X and measure p generate the cumulative distribution function (CDF) defined by

$$F(x) = \mu((-\infty, x]) = p(X \leq x)$$

where $p(X \leq x)$ is a short-hand notation for

$$p(\{\omega : \omega \in \Omega, X(\omega) \in (-\infty, x]\})$$

Theorem 3.3.1 A distribution function F for a random variable X uniquely defines the measure μ .

Proof is outside the scope of this class.

Example 3.3.2 Uniform Distribution

P is uniform on $[0, 1]$ and $X(\omega) = \omega$

For $x \in [0, 1]$

$$\begin{aligned} F(x) &= \mu((-\infty, x]) && \leftarrow \text{by definition of CDF} \\ &= p(X \leq x) && \leftarrow \text{by Thm 3.2 (definition of distribution)} \\ &= p(\{\omega : X(\omega) \in (-\infty, x]\}) \\ &= p((-\infty, x]) && \leftarrow \text{by } X(\omega) = \omega \\ &= p([0, x]) && \leftarrow \text{because } P \text{ is a uniform/Lebesgue measure} \\ &= x \end{aligned}$$

Theorem 3.4 Properties of Distribution Function

A distribution function satisfies the following properties:

- i) For all $x \leq y \in R, F(x) \leq F(y)$
- ii) $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow +\infty} F(x) = 1$
- iii) F is right-continuous proof:

- i) $x \leq y \in R$

$$F(x) = \mu((-\infty, x]), F(y) = \mu((-\infty, y])$$

$$\text{Since } x \leq y, \quad (-\infty, x] \subseteq (-\infty, y]$$

$$\mu((-\infty, x]) \leq \mu((-\infty, y])$$

"bigger set, bigger probability"

$$(A \subseteq B, \quad p(A) \leq p(B) \quad / \quad \mu(A) \leq \mu(B))$$

- ii) Define $A_n = \{X \leq -n\}, \quad n \geq 0$

Then $A_1 \supseteq A_2 \supseteq A_3 \supseteq \dots$

$$\begin{aligned}\bigcap_{n=1}^{\infty} A_n &= \phi \\ \lim_{n \rightarrow \infty} F(-n) &= \lim_{n \rightarrow \infty} p(A_n) \\ &= p\left(\bigcap_{n=1}^{\infty} A_n\right) \\ &= p(\phi) = 0\end{aligned}$$

Define $B_n = \{X \leq n\}$,
then $B_1 \subseteq B_2 \subseteq \dots$

$$\begin{aligned}\bigcup_{n=1}^{\infty} B_n &= \Omega \\ \lim_{n \rightarrow \infty} F(n) &= \lim_{n \rightarrow \infty} p(B_n) \\ &= p\left(\bigcup_{n=1}^{\infty} B_n\right) \\ &= p(\Omega) = 1\end{aligned}$$

iii) Fix $x \in R$ and let $h_n \downarrow 0$ ($h_n \geq 0$, $h_n \rightarrow 0$, $h_1 \geq h_2 \geq \dots$)
Define $C_n = \{X \leq x + h_n\}$
then $C_1 \supseteq C_2 \supseteq \dots$

$$\lim_{n \rightarrow \infty} C_n = \bigcap_{n=1}^{\infty} C_n = \{X \leq x\}$$

Therefore

$$\lim_{n \rightarrow \infty} F(x + h_n) = \lim_{n \rightarrow \infty} p(C_n) = p\left(\bigcap_{n=1}^{\infty} C_n\right) = p(X \leq x) = F(x)$$

Hence

$$\lim_{h \downarrow 0} F(x + h) = F(x)$$

So F is right-continuous.

Def. 3.5 The Inverse CDF

The inverse CDF of a r.v. X is defined by

$$F^{-1}(y) = \sup \{x : F(x) < y\}$$

Theorem 3.5.1 If F satisfies properties i) to iii) of Thm 3.4, then such an F is the CDF of some r.v.

Theorem 3.6 If F is a CDF, it can only have countably many discontinuities.

Lecture 4: Random Variables & Distributions

Sep 17, 2026

Def. 3.7 Density Function

A r.v. has a density function

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad \text{if for all } x \in \mathbb{R}$$
$$F(x) = \int_{-\infty}^x f(y) dy$$

we call $f(\cdot)$ the density function.

Def. 3.8 Joint Distribution

We define the joint distribution function of a random vector $\vec{X} = (X_1, \dots, X_n)$ by

$$F(x_1, \dots, x_n) = p(X_1 \leq x_1, \dots, X_n \leq x_n)$$

using the notation $p(A, B) \stackrel{\text{def}}{=} p(A \cap B)$

Theorem 3.8.1 A joint distribution $F : \mathbb{R}^n \rightarrow \mathbb{R}$ satisfies the following properties:

i) if $(x_1, \dots, x_n)$ and $x'_1, \dots, x'_n$ satisfy $x_i \leq x'_i$ for all $i \in [n]$.

$$F(x_1, \dots, x_n) \leq F(x'_1, \dots, x'_n)$$

ii) $\lim_{x_i \rightarrow -\infty} F(x_1, \dots, x_n) = 0$ for all $i \in \{1, \dots, n\}$ and $\lim_{x_1 \rightarrow \infty, \dots, x_n \rightarrow \infty} F(x_1, \dots, x_n) = 1$

iii) $\lim_{h \rightarrow 0^+} F(x_1 + h, \dots, x_n) = \dots = \lim_{h \rightarrow 0^+} F(x_1, \dots, x_n + h)$
 $= F(x_1, \dots, x_n)$

Proof are analogous to the proof under univariate case.

Def. 3.9 The Joint Density

A random vector has a joint density function:

$$f : \mathbb{R}^n \rightarrow \mathbb{R}, \quad \text{if } \forall (x_1, \dots, x_n) \in \mathbb{R}^n$$
$$F(x_1, \dots, x_n) = \int_{-\infty}^{x_1} \dots \int_{-\infty}^{x_n} f(y_1, \dots, y_n) dy_n \dots dy_1$$

Theorem 3.10 Independence of Random Variables

Random variables $X_1, \dots, X_n$ with CDFs $F_1, \dots, F_n$ are independent if and only if

$$F(x_1, \dots, x_n) = \prod_{i=1}^n F_i(x_i), \quad \forall (x_1, \dots, x_n) \in \mathbb{R}^n$$

Limits of sets:

Let A_i be sets for $i \in \{1, 2, \dots\}$ and consider $\lim_{n \rightarrow \infty} A_n$.

- If $A_1 \subseteq A_2 \subseteq \dots \subseteq A_n \subseteq \dots$ (increasing), then $\lim_{n \rightarrow \infty} \bigcup_{i=1}^n A_i = \bigcup_{i=1}^{\infty} A_i$.
- If $A_1 \supseteq A_2 \supseteq \dots \supseteq A_n \supseteq \dots$ (decreasing), then $\lim_{n \rightarrow \infty} \bigcap_{i=1}^n A_i = \bigcap_{i=1}^{\infty} A_i$.

Lecture 5:

Sep 22, 2026

Expectation

In STA237:

$$E(X) = \begin{cases} \sum_x x \cdot p(x) & \text{if } X \text{ discrete} \\ \int x f(x) dx & \text{if } X \text{ continuous} \end{cases}$$

for this course, we are defining probability via "measure theory", so we need to revisit the definition of expectation.

Under the "measure theory" perspective:

$$E(X) = \int x dP \quad \Rightarrow \quad P = \text{probability measure}$$
$$\int x f(x) dx \quad \Rightarrow \quad \int x dP$$

"Reinmann Integral"

Def. 5.1 Simple Expectation

If X only takes finitely many values $x_1, \dots, x_n$, and $A_i : \{\omega : X(\omega) = x_i\}$, define the expectation of X by:

$$E(X) = \sum_{i=1}^n x_i P(A_i)$$

Example 5.1.1 For an arbitrary set $A \subset \Omega$, let

$$Y(\omega) = I\{\omega \in A\} \quad \text{Then } E(Y) = P(A)$$

$$\begin{aligned} E(Y) &= \sum_{i=0}^1 y_i \cdot P(A_i) \quad Y \text{ only takes two values 0 and 1} \\ &= 1 \cdot P\{\omega \in A\} + 0 \cdot P\{\omega \notin A\} \\ &= P(A) = P(\{\omega : \omega \in A\}) \end{aligned}$$

Example 5.1.2 P is uniform measure on $\Omega = [0, 1]$, and

$$\begin{aligned} X(\omega) &= \begin{cases} 5 & \omega > \frac{1}{3} \\ 3 & \omega < \frac{1}{3} \end{cases} \\ E(X) &= 5 \cdot P(\omega > \frac{1}{3}) + 3 \cdot P(\omega < \frac{1}{3}) \\ &= 5 \cdot (1 - \frac{1}{3}) + 3 \cdot \frac{1}{3} \\ &= \frac{13}{3} \end{aligned}$$

Proposition 5.2 Properties of Expectation

If X and Y are simple random variables, the following hold:

- i) If $x \geq 0$ a.s., $E(X) \geq 0$
- ii) For any $a \in \mathbb{R}$, $E(aX) = a \cdot E(X)$
- iii) $E(X + Y) = E(X) + E(Y)$

Remark: "a.s." means almost surely. if E a.s. $\Rightarrow P(E) = 1$

$x \geq 0$ a.s. $\Rightarrow P(X \geq 0) = 1 \Leftrightarrow P(\{\omega : X(\omega) \geq 0\}) = 1$

we may have certain ω 's s.t. $X(\omega) < 0$, but $P(\{\omega : X(\omega) < 0\}) = 0$

Proof: Both i) and ii) are trivial.

For iii), let X take values $x_1, \dots, x_n$ on sets $A_1, \dots, A_n$ (i.e. $A_i = \{\omega : X(\omega) = x_i\}$) and Y takes values $y_1, \dots, y_m$ on sets $\beta_1, \dots, \beta_m$,

Observe that there are $m \cdot n$ events

$$(C_{ij} = \{X = x_i \cap Y = y_j\})$$

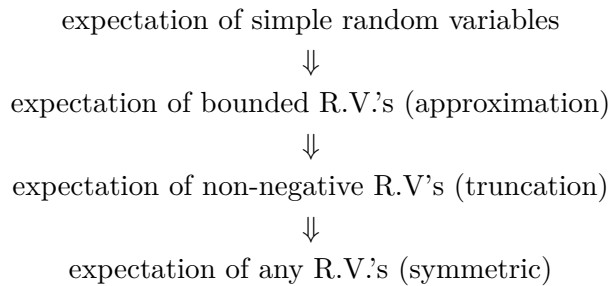
Then

$$\begin{aligned} E(X + Y) &= \sum_{i=1}^n \sum_{j=1}^m (x_i + y_j) \cdot P(C_{ij}) \\ &= \sum_{i=1}^n x_i \sum_{j=1}^m P(C_{ij}) + \sum_{j=1}^m y_j \sum_{i=1}^n P(C_{ij}) \\ &= E(X) + E(Y) \end{aligned}$$

Lemma 5.2.1 If Proposition 5.2 holds, the following properties also hold:

- iv) If $X \leq Y$ a.s., then $E(X) \leq E(Y)$
- v) If $X = Y$ a.s., then $E(X) = E(Y)$
- vi) $|E(X)| \leq E(|X|)$ proof as exercise.

Our logic is, we first define

**Def. 5.3 Bounded Random Variable**

A random variable X is bounded if there exists $m < \infty$, such that $|x| \leq m$ a.s.

Def. 5.3.1 Expectation of Bounded Random Variable

If X is a bounded R.V., then

$$\begin{aligned} E(X) &= \inf \{E(Z) : Z \text{ simple, } Z \geq X \text{ a.s.}\} \\ &= \sup \{E(Y) : Y \text{ simple, } Y \leq X \text{ a.s.}\} \end{aligned}$$

Recall, when we defined the integral:

the integral is defined as the limit value of sums of areas of tiny rectangles.

When it comes to Reinmann integral, we are no longer cutting X axis into tiny intervals, but cutting each set into tiny subsets and caculate the sum of x times the measure of each tiny subset.

$$\begin{aligned} E(X) &= \inf \{E(Z) : Z \text{ is simple and } Z \geq X \text{ a.s.}\} \\ &= \sup \{E(Y) : Y \text{ is simple and } Y \leq X \text{ a.s.}\} \end{aligned}$$

"Squeeze the R.V. X via two simple R.V.'s Y and Z , with one captures the upper bound and the others captures the lower bound".

Proof: Since X is bounded, $|x| < m$ a.s.

$$L = \sup \{EY : Y \text{ simple and } Y \leq X \text{ a.s.}\}$$

$$U = \inf \{EZ : Z \text{ simple and } Z \geq X \text{ a.s.}\}$$

$$L \leq U \text{ a.s. } Y \leq X \text{ a.s. and } Z \geq X \text{ a.s.}$$

Our goal is to prove $L \geq U \Rightarrow L = U \stackrel{def}{=} E(X)$

the way to cut $[-m, m]$ into tiny intervals.

for any arbitrary n , we define

$$E_k = \left\{ \omega : \frac{km}{n} \geq X(\omega) \geq \frac{(k-1)m}{n} \right\}$$

we then define two simple R.V.'s

Lower bound:

$$\phi_n(\omega) = \sum_k \frac{(k-1)m}{n} \cdot I(\omega \in E_k)$$

Upper bound:

$$\psi_n(\omega) = \sum_k \frac{km}{n} \cdot I(\omega \in E_k)$$

$$\phi_n(\omega) \leq X(\omega) \leq \psi_n(\omega)$$

$$E\phi_n(\omega) \leq E(X) \leq E\psi_n(\omega)$$

$$E\psi_n(\omega) = E\phi_n(\omega) + \frac{m}{n}$$

$$X \leq Y \text{ a.s.}$$

"step " $\omega : X(\omega) \leq Y(\omega)$

"step " $\{\omega : X(\omega) \leq Y(\omega)\}$

Def. 5.4 Non-negative Random Variable

If $X \geq 0$ a.s. ($P(\{\omega : X(\omega) \geq 0\}) = 1$)

define $E(X) = \sup \{E(Y) : Y \text{ bounded, } 0 \leq Y \leq X \text{ a.s.}\}$

Remark: Def 5.4. only provides a definition of $E(X)$ when X is non-negative, it does not guarantee $E(X) < \infty$.

Def. 5.5 Integrable Random Variables

For any random variable X , let $X^+(\omega) = \max \{X(\omega), 0\}$, and $X^- = \max \{-X(\omega), 0\}$.

Observe that both X^+ and X^- are non-negative and $X = X^+ - X^-$.

Then, if $E(|X|) = E(X^+) + E(X^-) < \infty$, we say R.V. X is integrable and define $E(X) = E(X^+) - E(X^-)$

1. $E|X| = E(X^+) + E(X^-) < \infty \Rightarrow X$ is integrable
2. $E(X) = E(X^+) - E(X^-)$

Proposition 5.6 Properties of Expected Values

If R.V. X is either bounded, non-negative or integrable. Proposition 5.2. always holds.

1. X and Y may not need to be simple, if they are bounded, non-negative or integrable, the 3 properties (prop 5.2) also hold.
2. Lemma 5.2.1 says, if prop 5.2 holds, the additional 3 properties also hold.
3. To sum up, if R.V.'s are bounded, non-negative or integrable, all the 6 properties hold.

Hint for Assignment 1. Q2:

E_1 and E_2 are independent.

E_1^c and E_2^c are independent.

if $E_1, E_2, \dots, E_n$, you want to prove $E_1^c, E_2^c, \dots, E_n^c$ are independent.

$$P(E_1^c \cap E_2^c \cap \dots \cap E_n^c) = \dots$$

you can flip 1 event at a time

first try to flip the first...

$$\begin{aligned} & P(E_1^c \cap E_2^c \cap \dots \cap E_n^c) \\ &= P(E_2^c \cap E_3^c \cap \dots \cap E_n^c) - P(E_1 \cap E_2^c \cap \dots \cap E_n^c) \\ &= \dots = (1 - P(E_1))P(E_2^c) \dots P(E_n^c) \end{aligned}$$

Then you flip the second one...

Via induction, complete the proofs

Lecture 6:

Sep 24, 2026

Theorem 5.7 Independence and Expectation

If X and Y are independent with $EX, EY < \infty$, then

$$E(XY) = (EX)(EY)$$

Proof:

First, consider both X and Y are simple, so that

$$X = \sum_{i=1}^n x_i I(A_i) \quad \text{and} \quad Y = \sum_{j=1}^m y_j I(B_j)$$

$$\begin{aligned} E(XY) &= E\left[\sum_{i=1}^n \sum_{j=1}^m x_i y_j \cdot \mathbf{1}_{\{A_i \cap B_j\}}\right] \\ &= \sum_{i=1}^n \sum_{j=1}^m x_i y_j P(x_i \in A_i, Y_j \in B_j) \\ &= \sum_{i=1}^n \sum_{j=1}^m x_i y_j P(\{\omega : X(\omega) \in A_i, Y(\omega) \in B_j\}) \\ &= \sum_{i=1}^n \sum_{j=1}^m x_i y_j P(X \in A_i) \cdot P(Y \in B_j) \\ &= \underbrace{\sum_{i=1}^n x_i \cdot P(X \in A_i)}_{EX} \underbrace{\sum_{j=1}^m y_j \cdot P(Y \in B_j)}_{EY} \\ &= (EX)(EY) \end{aligned}$$

Next, consider the case that X and Y are non-simple and non-negative, then we let $Z_1 \geq X, Z_2 \geq Y$, where Z_1 and Z_2 are simple. Notice that

$$Z_1 \geq X, Z_2 \geq Y \text{ implies that } Z_1 \cdot Z_2 \geq XY$$

$$\begin{aligned} (EX)(EY) &= \inf \{EZ_1 : Z_1 \text{ simple, } Z_1 \geq X \text{ a.s.}\} \\ &\quad \inf \{EZ_2 : Z_2 \text{ simple, } Z_2 \geq Y \text{ a.s.}\} \end{aligned}$$

$$\begin{aligned} &\inf \{(EZ_1) \cdot (EZ_2)\} \quad \text{because of independence and } Z_1, Z_2 \text{ simple} \\ &\geq \inf \{EZ_1 Z_2, Z_1 \text{ and } Z_2 \text{ simple and indep, } Z_1 Z_2 \geq XY \text{ a.s.}\} \end{aligned}$$

Additionally, if Z_1 and Z_2 are simple and

$$Z_1 \leq X \text{ a.s.} \quad Z_2 \leq Y \text{ a.s.}$$

Similarly, $(EX)(EY = \sup \{EZ_1Z_2, Z_1 \text{ and } Z_2 \text{ are simple and indep, } Z_1 \leq X, Z_2 \leq Y \text{ a.s.}\})$

Previously, we proved that

$$\inf \{EZ_1Z_2, \quad Z_1, Z_2 \text{ are simple and indep with } Z_1 \geq X, Z_2 \geq Y \text{ a.s.} \} = E(XY)$$

and

$$\sup \{EZ_1Z_2, \quad Z_1, Z_2 \text{ are simple and indep with } Z_1 \leq X, Z_2 \leq Y \text{ a.s.} \} = E(XY)$$

Finally, for X and Y that are integrable, first realize that if X and Y are independent, then X^+, X^-, Y^+, Y^- are also independent.

$$X = X^+ - X^- \quad Y = Y^+ - Y^-$$

$$\begin{aligned} E(XY) &= E[(X^+ - X^-)(Y^+ - Y^-)] \\ &= E[X^+Y^+ - X^+Y^- - X^-Y^+ + X^-Y^-] \\ &= E(X^+Y^+) - E(X^+Y^-) - E(X^-Y^+) + E(X^-Y^-) \\ &= E(X^+)E(Y^+) - E(X^+)E(Y^-) - E(X^-)E(Y^+) + E(X^-)E(Y^-) \\ &= (EX)(EY) \end{aligned}$$

Def. 5.8 Variance and Covariance

The variance of a random variable X is defined as

$$\text{Var}(X) = E[(X - EX)^2]$$

The covariance of random variables X and Y is

$$\text{Cov}(X, Y) = E[(X - EX)(Y - EY)]$$

The correlation coefficient is

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}}$$

Lecture 7: Chapter 4 - Inequalities

Sep 29, 2026

Theorem 4.1 Markov's Inequality

If $X \geq 0$ a.s., then for all $a > 0$,

$$P(X \geq a) \leq \frac{E(X)}{a}$$

Proof:

Define $Z(\omega) = a \cdot I\{X(\omega) \geq a\}$. Then, $Z \leq X$ a.s., and since Z is simple,

$$E(Z) = a \cdot P(X \geq a)$$

$$P(X \geq a) = \frac{E(Z)}{a} \leq \frac{E(X)}{a}$$

Theorem 4.2 Chebyshev's Inequality

For all $t \geq 0$,

$$P(|X - E(X)| \geq t) \leq \frac{\text{Var}(X)}{t^2}$$

Proof:

Apply Markov's inequality to $Y = [X - E(X)]^2$

Both Chebyshev and Markov's inequality talk about

$$P(X \geq \dots) \quad \Leftarrow \quad \text{the tail probability}$$

"Concentration of Probability Mass"

Where Markov uses the 1st moment, Chebyshev uses 2nd moment.

What if we use other transformation/moments to quantify the tail probability?

E.g. the exponential moment? $\Rightarrow$ Chernoff's Inequality

$$M_X(t) = E[e^{tx}]$$

Theorem 4.3 Chernoff's Inequality

For all $t \geq 0$, $P(X \geq E(X) + t) \leq \inf_{\lambda > 0} M_X - E(X)(\lambda)e^{-\lambda t}$

$$P(X \geq E(X) + t) \text{ roughly } P(\underbrace{X - E(X)}_{\substack{\text{distance between } X \\ \text{to its expected value}}} \geq t)$$

$$\leq \inf_{\lambda > 0} M_{X-E(X)}(\lambda)e^{-\lambda t}$$

Proof:

the function $g(x) = e^x$ is monotonically increasing and non-negative.

$$\begin{aligned} P(X - E(X) \geq t) \\ &= P(e^{X-E(X)} \geq e^t) \\ &= P(e^{\lambda(X-E(X))} \geq e^{\lambda t}) \quad \lambda \geq 0 \end{aligned}$$

Since $e^{\lambda(X-E(X))} \geq 0$ a.s., we can apply Markov's inequality

$$\leq \frac{E[e^{\lambda(X-E(X))}]}{e^{\lambda t}} \quad \text{where the num. is the } m_{gf} \quad M_{X-E(X)}$$

We proved that

$$P(X - E(X) \geq t) \leq e^{-\lambda t} \cdot M_{X-E(X)}(\lambda)$$

Since this inequality holds for any arbitrary $\lambda \geq 0$, we can take the inf:

$$P(X - E(X) \geq t) \leq \inf_{\lambda > 0} e^{-\lambda t} M_{X-E(X)}(\lambda)$$

Theorem 4.4 Hoeffding's Inequality

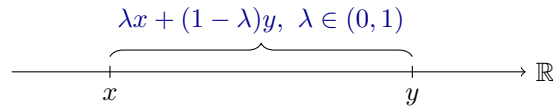
Def. 4.4.1 Convex Function

A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is convex if for all $\lambda \in (0, 1)$ and $x, y \in \mathbb{R}$

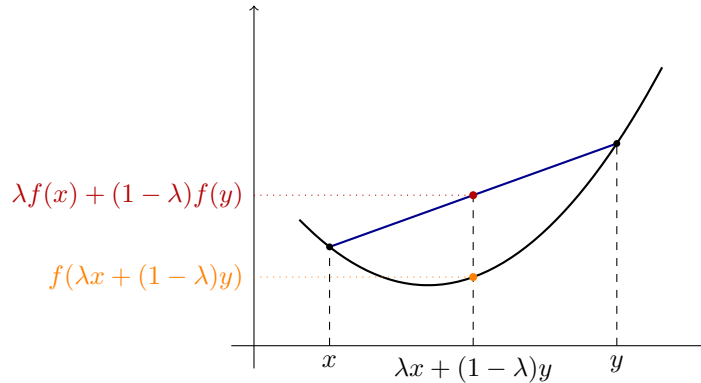
$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$$

Any value between x and y can be represented as

$$\lambda x + (1 - \lambda)y \text{ for } \lambda \in (0, 1)$$



if $\lambda \rightarrow 1$, then the value is closer to x . Vice versa for $\lambda \rightarrow 0$, the value is closer to y .



Convexity: $f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$

$\Rightarrow$ *the function is under the straight line*

Lemma 4.4.2 Hoeffding's Lemma

If Y satisfies $E(Y) = 0$ and $a \leq Y \leq b$ a.s., then for all $\lambda > 0$

$$M_Y(\lambda) \leq e^{\lambda^2(b-a)^2/8}$$

Proof:

Since $a \leq Y \leq b$ a.s., $Y = \alpha \cdot a + (1 - \alpha) \cdot b$.

Transforming $Y \Rightarrow \alpha$, where $0 \leq \alpha \leq 1$,

$$\alpha = \frac{b - Y}{b - a}, \quad 1 - \alpha = \frac{Y - a}{b - a}$$

Also, since the function $e^{\lambda y}$ is convex,

$$e^{\lambda(\alpha a + (1-\alpha)b)} \leq \alpha e^{\lambda a} + (1 - \alpha)e^{\lambda b}$$

Apply $f(y) = e^{\lambda y}$ transformation to r.v. Y :

$$e^{\lambda Y} \leq \alpha e^{\lambda a} + (1 - \alpha)e^{\lambda b} = \frac{b - Y}{b - a} e^{\lambda a} + \frac{Y - a}{b - a} e^{\lambda b}$$

Take expectation on both sides:

$$\begin{aligned} E[e^{\lambda Y}] &\leq E\left[\frac{b - \overbrace{Y}^{\text{random var.}}}{b - a} \underbrace{e^{\lambda a}}_{\text{constant}}\right] + E\left[\frac{Y - a}{b - a} \underbrace{e^{\lambda b}}_{\text{constant}}\right] \\ &= \frac{b - E(Y)}{b - a} e^{\lambda a} + \frac{E(Y) - a}{b - a} e^{\lambda b} \\ &= \underbrace{\frac{b}{b - a}}_p e^{\lambda a} + \frac{-a}{b - a} e^{\lambda b} \quad (\text{since } E(Y) = 0) \end{aligned}$$

Next, we let $p = \frac{b}{b-a}$:

$$E[e^{\lambda Y}] = p e^{\lambda a} + (1 - p) e^{\lambda b}$$

We take log on RHS:

$$\begin{aligned} \log\{E[e^{\lambda Y}]\} &\leq \log\{p e^{\lambda a} + (1 - p) e^{\lambda b}\} \\ &= \log\{e^{\lambda a} [p + (1 - p) e^{\lambda(b-a)}]\} \\ &= \underbrace{\lambda a}_{=(p-1)u} + \log[p + (1 - p) e^{\lambda(b-a)}] \\ (p - 1)u &= \left(\frac{b}{b - a} - 1\right) (b - a)\lambda = \frac{a}{b - a} (b - a)\lambda = a\lambda \end{aligned}$$

Next, let $u = (b - a)\lambda$:

$$= (p - 1)u + \log[p + (1 - p)e^u] = \varphi(u)$$

$$\varphi(0) = 0, \quad \varphi'(u) = (p - 1) + \frac{(1 - p)e^u}{p + (1 - p)e^u}, \quad \varphi'(0) = 0$$

$$\varphi(u) = \underbrace{\varphi(0)}_0 + \underbrace{\varphi'(0)}_0 u + \frac{\varphi''(\xi)}{2} u^2 + \dots \quad (\text{Taylor's Expansion at } u = 0)$$

For some ξ between 0 and u , $\varphi(u) = \frac{1}{2}\varphi''(\xi)u^2$

$$\varphi''(u) = \frac{p(1 - p)e^u}{[p + (1 - p)e^u]^2} \leq \frac{1}{4}$$

$$\Rightarrow \varphi(u) \leq \frac{1}{8}u^2$$

$$\Rightarrow \log\{E[e^{\lambda Y}]\} \leq \frac{1}{8}u^2 = \frac{(b - a)^2 \lambda^2}{8} \quad \text{as } u = (b - a)\lambda$$

$$\Rightarrow M_Y(\lambda) \leq e^{\frac{(b-a)^2 \lambda^2}{8}}$$

□

Hoeffding's Inequality:

Suppose $X_1, X_2, \dots, X_n, \dots$ are indep. with $a_i \leq X_i \leq b_i$ for all i .

Then letting $S_n = \sum_{i=1}^n X_i$, for all $t > 0$,

$$P(|S_n - E(S_n)| \geq t) \leq 2 \cdot e^{-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}}$$

Proof:

We first prove the $P(S_n - E(S_n) \geq t)$ part.

Applying Chernoff's inequality,

$$P(S_n - E(S_n) \geq t) \leq e^{-\lambda t} E[e^{\lambda(S_n - E(S_n))}]$$

$$S_n - E(S_n) = \sum_{i=1}^n (X_i - E(X_i)), \quad S_n = \sum_{i=1}^n X_i$$

$$e^{\lambda(S_n - E(S_n))} = e^{\lambda \sum_{i=1}^n (X_i - E(X_i))} = \prod_{i=1}^n \underbrace{e^{\lambda(X_i - E(X_i))}}_{M_{X_i - E(X_i)}(\lambda)}$$

$$M_{S_n - E(S_n)}(\lambda) = \prod_{i=1}^n M_{X_i - E(X_i)}(\lambda)$$

Applying Hoeffding's Lemma: $M_{X_i - E(X_i)}(\lambda) \leq e^{\frac{(b_i - a_i)^2 \lambda^2}{8}}$

$$M_{S_n - E(S_n)}(\lambda) \leq \prod_{i=1}^n e^{\frac{(b_i - a_i)^2 \lambda^2}{8}} = e^{\frac{\lambda^2}{8} \sum_{i=1}^n (b_i - a_i)^2}$$

Combine with Chernoff's inequality:

$$P(S_n - E(S_n) \geq t) \leq e^{-\lambda t + \frac{\lambda^2}{8} \sum_{i=1}^n (b_i - a_i)^2}$$

Rearrange, we have Hoeffding's inequality.

□